Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The total weight of a piece of wood is $6\ \text{kg}$. In the floating state in water its $\frac{1}{3}$ part remains inside the water. On this floating piece of wood what maximum weight is to be put such that the whole of the piece of wood is to be drowned in the water?

(A) $15\ \text{kg}$

(B) $14\ \text{kg}$

(C) $10\ \text{kg}$

(D) $12\ \text{kg}$

Q.2

A right circular cone of density ρ, floats just immersed with its vertex downwards in a vessel containing two liquids of densities σ_1 and σ_2 respectively, the planes of separation of the two liquids cuts off from the axis of the cone a fraction z of its length. Find z.

(A) $\left(\frac{\rho+\sigma_2}{\sigma_1+\sigma_2}\right)^{1/3}$

(B) $\left(\frac{\rho-\sigma_2}{\sigma_1-\sigma_2}\right)^{1/3}$

(C) $\left(\frac{\rho-\sigma_2}{\sigma_1+\sigma_2}\right)^{1/2}$

(D) $\left(\frac{\rho-\sigma_2}{\sigma_1-\sigma_2}\right)$

Q.3

The pressure at the bottom of a tank containing
liquid does not depend on

(A) acceleration due to gravity

(B) height of the liquid column

(C) area of the bottom surface

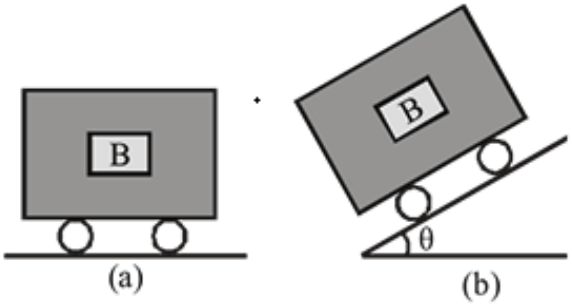
(D) nature of the liquid

Q.4

A body B is capable of remaining stationary inside a liquid at the position shown in Fig. (a). If the whole

@IITJEE_Advanced

system is gently placed on smooth inclined plane (Fig (b)) and is allowed to slide down, then $(0 < \theta < 90^\circ)$. The body will



(A) move up (relative to liquid)

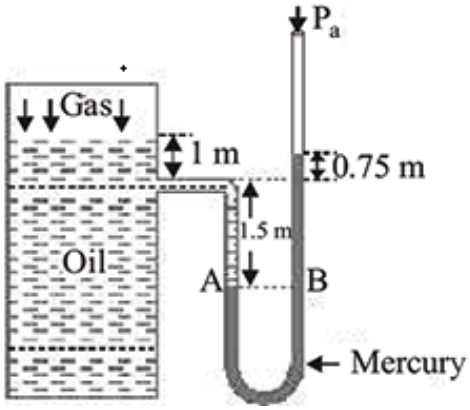
(B) move down (relative to liquid)

(C) remain stationary (relative to liquid)

(D) move up for some inclination θ and will move down for another inclination θ

Q.5

What is the absolute pressure of the gas above the liquid surface in the tank shown in fig. Density of oil $=$ $820\ \mathrm{kg/m^3}$, density of mercury $= 13.6 \times 10^3\ \mathrm{kg/m^3}$
Given 1 atmospheric pressure $= 1.01 \times 10^5\ \mathrm{N/m^2}$



(A) $3.81 \times 10^5\ \mathrm{N/m^2}$

(B) $6 \times 10^6\ \mathrm{N/m^2}$

(C) $5 \times 10^7\ \mathrm{N/m^2}$

(D) $4.6 \times 10^2\ \mathrm{N/m^2}$

Q.6

The two thigh bones, each of cross-sectional area $10\ \mathrm{cm^2}$ support the upper part of a human body of mass $40\ \mathrm{kg}$. Estimate the average pressure sustained by the bones. Take $g = 10\ \mathrm{m/s^2}$

(A) $2 \times 10^5\ \mathrm{N/m^2}$

(B) $5 \times 10^4\ \mathrm{N/m^2}$

(C) $2 \times 10^7\ \mathrm{N/m^2}$

@IITJEE_Advanced

(D) $3 \times 10^6\ \mathrm{N/m^2}$

Q.7

A boy can reduce the pressure in his lungs to $750\ \mathrm{mm}$ of mercury. Using a straw he can drink water from a glass upto the maximum depth of (atmospheric pressure $= 760\ \mathrm{mm}$ of mercury; density of mercury $= 13.6\mathrm{gcm^{-3}}$)

(A) $13.6\ \mathrm{cm}$

(B) $9.8\ \mathrm{cm}$

(C) $10\ \mathrm{cm}$

(D) $76\ \mathrm{cm}$

Q.8

The density ρ of water of bulk modulus B at a depth y in the ocean is related to the density at surface ρ_0 by the relation

(A) $\rho = \rho_0 \left[1 - \frac{\rho_0 gy}{B}\right]$

(B) $\rho = \rho_0 \left[1 + \frac{\rho_0 gy}{B}\right]$

(C) $\rho = \rho_0 \left[1 + \frac{B}{\rho_0 hgy}\right]$

(D) $\rho = \rho_0 \left[1 - \frac{B}{\rho_0 hgy}\right]$

Q.9

Two non-mixing liquids of densities ρ and $n\rho\ (\mathrm{n} > 1)$ are put in a container. The height of each liquid is h. A solid cylinder of length Land density d is put in this container. The cylinder floats with its axis vertical and length $\mathrm{pL}(\mathrm{p} < 1)$ in the denser liquid. The density d is equal to :

(A) $\{1 + (\mathrm{n}+1)\mathrm{p}\}\rho$

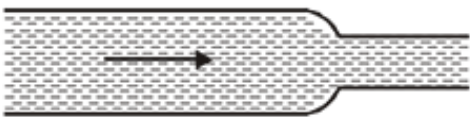
(B) $\{2 + (n+1)p\}\rho$

(C) $\{2 + (\mathrm{n}-1)\mathrm{p}\}\rho$

(D) $\{1 + (\mathrm{n}-1)\mathrm{p}\}\rho$

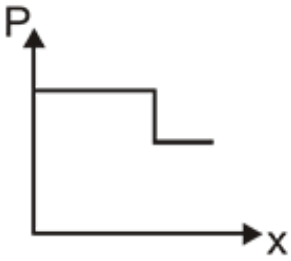
Q.10

Water flows through a frictionless tube with a varying cross-section as shown in fig. Pressure P at points along the axis is represented by

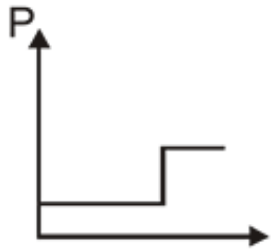


@IITJEE_Advanced

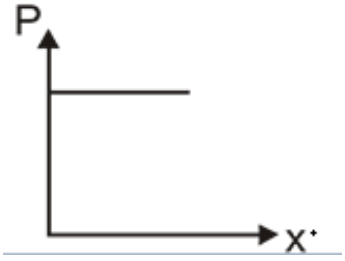
(A)



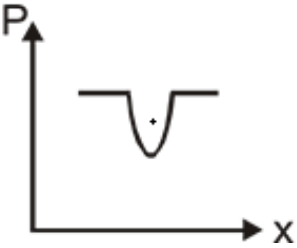
(B)



(C)



(D)



Q.11

An aquarium is filled with water. The lateral wall of the aquarium is 50 cm long and 40 cm high. Using 10 m s^{-2} for the acceleration due to gravity, and 1 g cm^{-3} for the density of water, the force on the lateral wall of the aquarium is

(A) 36 N

(B) 400 N

(C) 180 N

(D) 1500 N

Q.12

A sphere of solid material of specific gravity 8 has a concentric spherical cavity and just sinks in water. The ratio of radius of cavity to that of outer radius of the sphere must be

(A) $\frac{7^{1/3}}{2}$

(B) $\frac{5^{1/3}}{2}$

(C) $\frac{9^{1/3}}{2}$

(D) $\frac{3^{1/3}}{2}$

@IITJEE_Advanced

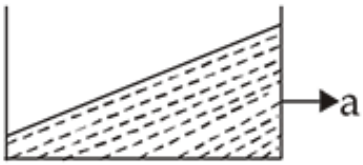
Q.13

A vessel containing water is given a constant acceleration ' a ' towards the right along a straight horizontal path. Which of the following diagrams represents the surface of the liquid?

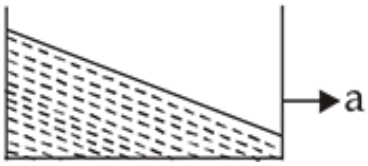
(A)



(B)



(C)



(D) None of these

Q.14

A horizontal tube has different cross sections at points A and B. The area of cross section are a_1 and a_2, respectively, and pressures at these points are $\mathrm{p}_1 = \rho \mathrm{g} \mathrm{h}_1$ and $\mathrm{p}_2 = \rho \mathrm{g} \mathrm{h}_2$, where ρ is the density of liquid flowing in the tube and h_1 and h_2 are heights of liquid columns in vertical tubes connected at A and B. If $\mathrm{h}_1 - \mathrm{h}_2 = \mathrm{h}$, then the flow rate of the liquid in the horizontal tube is

(A) $a_1 a_2 \sqrt{\frac{2gh}{a_1^2 - a_2^2}}$

(B) $a_1 a_2 \sqrt{\frac{2g}{h\left(a_1^2 - a_2^2\right)}}$

(C) $a_1 a_2 \sqrt{\frac{\left(a_1^2 + a_2^2\right)h}{2g\left(a_1^2 - a_2^2\right)}}$

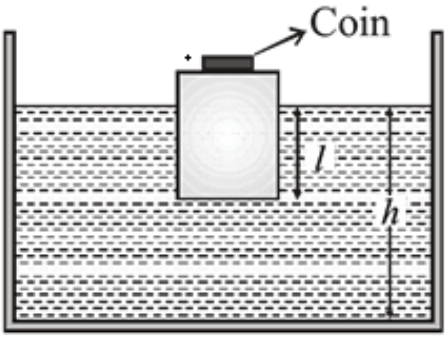
(D) $\frac{2\mathrm{a}_1 \mathrm{a}_2 \mathrm{gh}}{\sqrt{\mathrm{a}_1^2 - \mathrm{a}_2^2}}$

Q.15

A wooden block, with a coin placed on its top, floats in water as shown in fig. the distance l and h are shown

@IITJEE_Advanced

there. After some time the coin falls into the water. Then



(A) ℓ decreases and h increases

(B) ℓ increases and h decreases

(C) both ℓ and h increases

(D) both ℓ and h decreases

Q.16

In a hydraulic press, the small cylinder has a diameter of $d_1{}^1$ cm, while the large piston has diameter of $' \mathrm{d}_2{}' \mathrm{cm}$. If a force F_1 ' is applied to a small piston, the force on the large piston P_2 ' is given by

(A) $\mathbf{F}_2 = \frac{\mathrm{d}_2^2}{\mathrm{d}_1^2} \mathrm{F}_1$

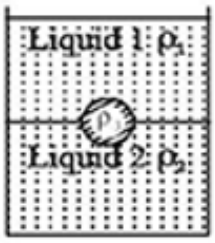
(B) $\mathrm{F}_2 = \frac{d_1^2}{d_2^2} F_1$

(C) $\mathrm{F}_2 = \frac{\mathrm{d}_1^2}{\mathrm{d}_1^2} \frac{1}{\mathrm{F}_\mathrm{i}}$

(D) $\mathrm{F}_2 = \frac{d_2^2}{d_1^2} \frac{1}{F_1}$

Q.17

A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and, ρ_2 respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1, ρ_2 and ρ_3?



(A) $\rho_3 < \rho_1 < \rho_2$

(B) $\rho_1 > \rho_3 > \rho_2$

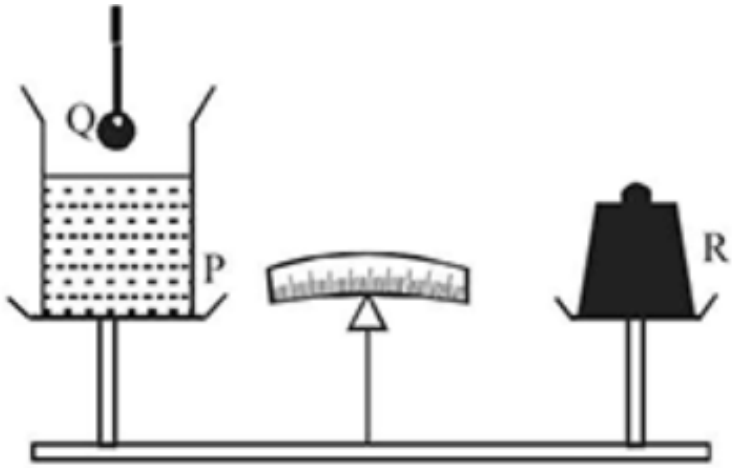
(C) $\rho_1 < \rho_2 < \rho_3$

(D) $\rho_1 < \rho_3 < \rho_2$

@IITJEE_Advanced

Q.18

Figure shows a weigh-bridge, with a beaker P with water on one pan and a balancing weight R on the other. A solid ball Q is hanging with a thread outside water. It has volume 40 cm^3 and weighs $80\ \mathrm{g}$. If this solid is lowered to sink fully in water, but not touching the beaker anywhere, the balancing weight R' will be



(A) same as R

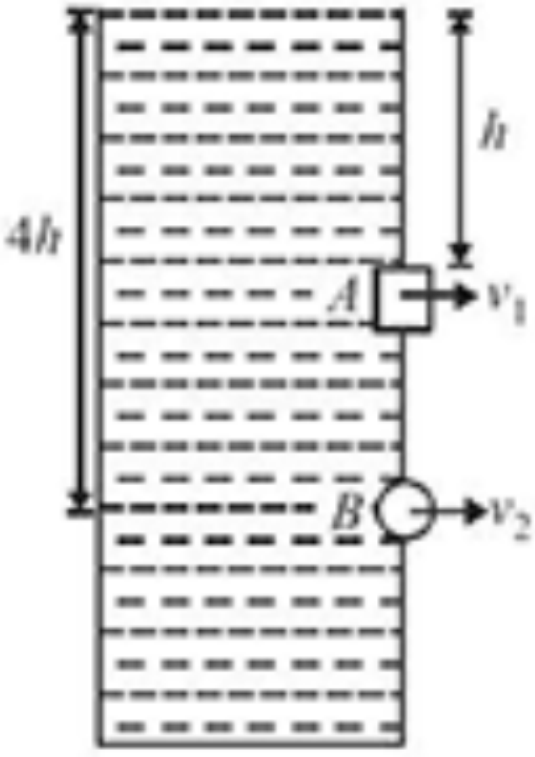
(B) $40\ \mathrm{g}$ less than R

(C) $40\ \mathrm{g}$ more than R

(D) $80\ \mathrm{g}$ more than R

Q.19

A square hole of side length ℓ is made at a depth of h and a circular hole of radius r is made at a depth of $4\ \mathrm{h}$ from the surface of water in a water tank kept on a horizontal surface. If $\ell << h, r << h$ and the rate of water flow from the holes is the same, then r is equal to



@IITJEE_Advanced

(A) $\frac{\ell}{\sqrt{2\pi}}$

(B) $\frac{\ell}{\sqrt{3\pi}}$

(C) $\frac{\ell}{3\pi}$

(D) $\frac{\ell}{2\pi}$

Q.20

Consider a narrow glass U-tube with the inner radii of the limbs being 2 mm & 1 mm. Now, the tube is partially filled water whose surface tension is 0.07 $\mathrm{N/m^2}$, such that the difference in heights of water in limbs is Δh mm. What is $\Delta h = ?$

(A) 07.05

(B) 06.00

(C) 06.05

(D) 07.00

Answers & Solutions

Q.1 Answer:

12 kg

Solution:

Weight of submerged part of the block $\mathrm{W} = \frac{1}{3}\mathrm{v}($ **Density of water** $)\mathrm{g}$.

2

Excess weight, = weight of water having $\frac{2}{3}$ **volume of the block.** $\mathrm{W}' = \frac{2}{3}\mathrm{v}($ **Density of water** $)\mathrm{g}$

Dividing (ii) by (i), $\frac{\mathrm{W}'}{\mathrm{W}} = \frac{2/3}{1/3}$

$\therefore \mathrm{W}' = 2\ \mathrm{W} \Rightarrow \mathrm{W}' = 2 \times 6 = 12$ kg

Q.2 Answer:

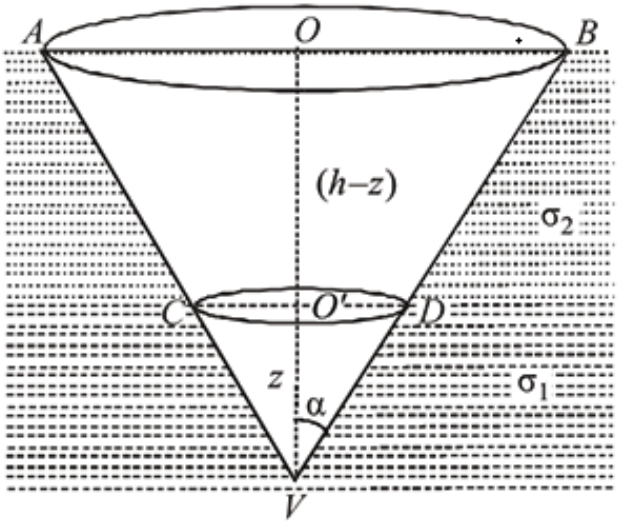
$\left(\frac{\rho-\sigma_2}{\sigma_1-\sigma_2}\right)^{1/3}$

Solution:

VAB **is the given cone. Let its height be** h **and semi-vertical angle** α**. Let the base** AB **of the cone be in the surface.** CD **is the surface of separation of two liquids,** O **and** O' **are the centres of the**

@IITJEE_Advanced

base AB and surface of separation CD.



$\therefore$ For equilibrium, weight of the cone $=$ (weight of liquid of density σ_1 displaced) $+$ (weight of liquid of density σ_2 displaced)

or $\frac{1}{3}\pi h^3 \tan^2\alpha\rho g = \frac{1}{3}\pi z^3 \tan^2\alpha\sigma_1 g$

$+\frac{1}{3}\pi\left(h^3 - z^3\right)\tan^2\alpha\cdot\sigma_2 g$

or $h^3\rho = z^3\sigma_1 + \left(h^3 - z^3\right)\sigma_2$

or $h^3\left(\rho - \sigma_2\right) = z^3\left(\sigma_1 - \sigma_2\right)$

or $z = h\left(\frac{\rho-\sigma_2}{\sigma_1-\sigma_2}\right)^{1/3}$.

Q.3 Answer:

area of the bottom surface

Solution:

$\mathrm{P} =$ **hpg** **i.e. pressure does not depend upon the area of bottom surface.**

Q.4 Answer:

move up for some inclination θ and will move down for another inclination θ

Solution:

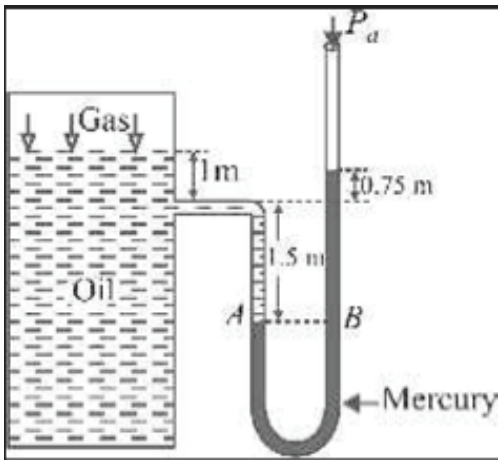
Due to acceleration down the plane, both body B and liquid will received the same component of acceleration down the plane.

Q.5 Answer:

$3.81 \times 10^5\ \mathrm{N/m^2}$

Solution:

@IITJEE_Advanced



Suppose P_{gas} is the pressure of the gas on the oil. As the points A and B are at the same level in the mercury columns, so $P_A = P_B$

or $P_{\text{gas}} + \rho_{\text{oil}}\, \text{gh}_{\text{oil}} = P_a + \rho_{\text{Hg}} g h_{\text{Hg}}$

or $P_{\text{gas}} + 820 \times 9.8 \times (1 + 1.50)$

$= P_a + 13.6 \times 10^3 \times 9.8 \times (1.5 + 0.75)$

or $P_{\text{gas}} + 20.09 \times 10^3 = P_{\text{a}} + 299.88 \times 10^3$

$\therefore P_{\text{gas}}^{\text{gas}} - Pa = 299.88 \times 10^3 - 20.09 \times 10^3$

or $[P_{\text{gas}}]_{\text{gauge}} = 279.8 \times 10^3\ \text{N/m}^2 = 2.8 \times 10^5\ \text{N/m}^2$

Absolute pressure of gas $[P_{\text{gas}}]_{\text{absolute}} = [P_{\text{gas}}]_{\text{gauge}} + P_a$

$= 2.8 \times 10^5 + 1.01 \times 10^5 = 3.81 \times 10^5\ \text{N/m}^2$

Q.6 Answer:

$2 \times 10^5\ \text{N/m}^2$

Solution:

Total cross-sectional area of the thigh bones $A = 2\left(10 \times 10^{-4}\right) = 2 \times 10^{-3}\ \text{m}^2$

Force acting on the bones $= mg = 40 \times 10 = 400\ \text{N}$

$\therefore \quad P_{\text{av}} = \frac{F}{A} = \frac{400}{2 \times 10^{-3}} = 2 \times 10^5\ \text{N/m}^2$

Q.7 Answer:

$13.6\ \text{cm}$

Solution:

Pressure difference between lungs and atmosphere $= (760 - 750)\text{mm of Hg}$

$= 10\ \text{mm of Hg} = 1\ \text{cm of Hg}$

Let the boy can suck water from depth h**. Then Pressure difference** $= h\rho_{\text{water}}\, g = 1\ \text{cm of Hg}$

or, $h \times 1\ \text{g cm}^{-3} \times 980\ \text{cm s}^{-2}$

$= 1\ \text{cm} \times 13.6\ \text{g cm}^{-3} \times 980\ \text{cm s}^{-2}$

$\therefore \quad h = 13.6\ \text{cm}$

The boy can suck water from the depth of $13.6\ \text{cm}$

@IITJEE_Advanced

Q.8 Answer:

$\rho = \rho_0 \left[1 + \frac{\rho_0 g y}{B}\right]$

Solution:

Bulk modulus, $B = -V_0 \frac{\Delta p}{\Delta V} \Rightarrow \Delta V = -V_0 \frac{\Delta p}{B}$

$\Rightarrow V = V_0 \left[1 - \frac{\Delta p}{B}\right]$

$\therefore$ **Density,** $\rho = \rho_0 \left[1 - \frac{\Delta p}{B}\right]^{-1} = \rho_0 \left[1 + \frac{\Delta p}{B}\right]$

where, $\Delta p = p - p_0 = h\rho_0 g$

= pressure difference between depth and surface of ocean $\therefore \rho = \rho_0 \left[1 + \frac{\rho_0 g y}{B}\right]$ (As $\mathrm{h} = \mathrm{y}$)

Q.9 Answer:

$\{1 + (\mathrm{n} - 1)\mathrm{p}\}\rho$

Solution:

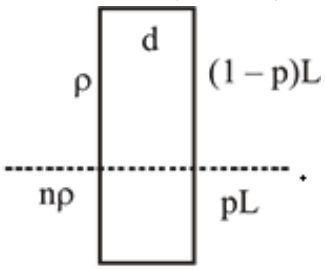
As we know, Pressure $\mathrm{P} = \mathrm{Vdg}$

Here, $\mathrm{LAdg} = (\mathrm{pL})\mathrm{A}(\mathrm{n}\rho)\mathrm{g}$

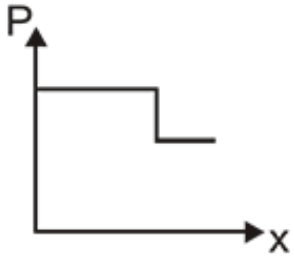
$+ (1 - \mathrm{p})\mathrm{LA}\rho\mathrm{g}$

nn

$\Rightarrow \quad \mathrm{d} = (1 - \mathrm{p})\rho + \mathrm{pn}\,\rho = [1 + (\mathrm{n} - 1)\mathrm{p}]\rho$



Q.10 Answer:



Solution:

Velocity of water is constant; lets say v_1 **in wider cross section. as water reaches from wider cross section to narrow cross section, its velocity increases all of a sudden. After that it again becomes constant, lets say** v_2**. So we have** $\mathrm{v}_2 > \mathrm{v}_1$**. According to Bernoulli's theorem,**

$P + \frac{1}{2}\rho v^2 = a$ constant

$\Rightarrow \mathrm{P}_2 < \mathrm{P}_1 \because \mathrm{v}_2 > \mathrm{v}_1$

$\Rightarrow$ option **A** is correct.

Q.11 Answer:

@IITJEE_Advanced

400 N

Solution:

Let us take the small strip having width of $\mathrm{d}h$ at depth h. Then, the pressure at the strip will be equal to, $P = \rho g h$, where ρ is the density, h is the depth of the liquid and g is acceleration due to gravity.

Then, the force on the strip will be equal to

$\mathrm{d}F = Pl\ \mathrm{d}h = \rho g h \times 0.5 \times \mathrm{d}h$

$\Rightarrow \int \mathrm{d}F = 0.5\rho g \int_0^{0.4} h\ \mathrm{d}h = 1000 \times 10 \times 0.5 \times \frac{0.4 \times 0.4}{2}$

$\Rightarrow F = 400\ \mathrm{N}$

Q.12 Answer:

$\frac{7^{1/3}}{2}$

Solution:

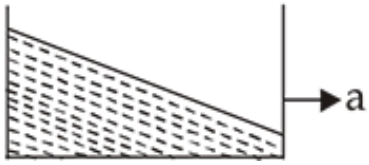
Let ρ be the density of the material. ρ_0 be the density of water when the sphere has just started sinking, the weight of the sphere = weight of water displaced (approx).

$\Rightarrow \frac{4}{3}\pi\left(\mathrm{R}^3 - \mathrm{r}^3\right)\rho \mathrm{g} = \frac{4}{3}\pi \mathrm{R}^3 \rho_0\ \mathrm{g}$

$\Rightarrow \left(\mathrm{R}^3 - \mathrm{r}^3\right)\rho = \mathrm{R}^3 \rho 0 \Rightarrow \frac{\left(\mathrm{R}^3 - \mathrm{r}^3\right)}{\mathrm{R}^3} = \frac{\rho^0}{\rho} = \frac{1}{8} \Rightarrow 1 - \frac{r^3}{R^3} = \frac{1}{8} \Rightarrow \frac{r^3}{R^3} = \frac{7}{8}$

$\Rightarrow \frac{r}{R} = \frac{(7)^{1/3}}{2}$

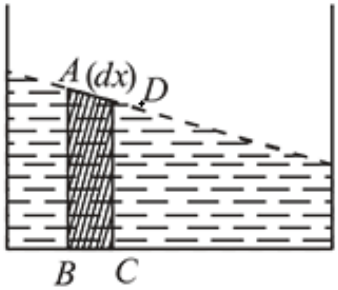
Q.13 Answer:



Solution:

Let us consider a small dotted segment of thickness dx for observation.

Since, this segment is accelerated towards right, a net force is acting in this segment towards right from the liquid towards the left of $ABCD$. According to Newton's third law, the segment $ABCD$ will also apply a force on the previous section creating a pressure on it which makes the liquid rise.



Q.14 Answer:

$a_1 a_2 \sqrt{\frac{2gh}{a_1^2 - a_2^2}}$

@IITJEE_Advanced

Solution:

Let a_1 and a_2 be cross sectional areas and v_1 and v_2 the velocities of liquid flow at A and B. If p_1 and p_2 are pressures of liquid recorded by manometer, then $p_1 + \frac{1}{2}pv_1^2 = p_2 + \frac{1}{2}\rho v_2^2$

$\Rightarrow p_1 - p_2 = \frac{1}{2}\rho v_1^2\left(\frac{v_2^2}{v_1^2} - 1\right)$

By equation of continuity, $a_1 v_1 = a_2 v_2$

Also, $p_1 - p_2 = hpg$

Substituting these values, we have $v_1 = \sqrt{\frac{2gh}{\frac{a_1^2}{a_2^2} - 1}}$

Rate of flow of liquid is $a_1 v_1 = a_1 a_2 \sqrt{\frac{2gh}{a_1^2 - a_2^2}}$

Q.15 Answer:

both ℓ and h decreases

Solution:

As the block moves up with the fall of coll, l decreases, similarly h will also decrease because when the coin is in water, it displaces water equal to its own volume only.

Q.16 Answer:

$\mathbf{F}_2 = \frac{d_2^2}{d_1^2} F_1$

Solution:

Using Pascal's law

$P_1 = \rho_2 \Rightarrow \frac{F_1}{\left(\frac{\pi d_1^2}{4}\right)} = \frac{F_2}{\left(\frac{\pi d_2^2}{4}\right)} \Rightarrow F_2 = \frac{d_2^2}{d_1^2} F_1$

Q.17 Answer:

$\rho_1 < \rho_3 < \rho_2$

Solution:

From the figure it is clear that liquid 1 floats on liquid 2 . The lighter liquid floats over heavier liquid. Therefore we can conclude that $\rho_1 < \rho_2$ Also $\rho_3 < \rho_2$ otherwise the ball would have sink to the bottom of the jar. Also $\rho_3 > \rho_1$ otherwise the ball would have floated in liquid 1 . From the above discussion we conclude that

$$\rho_1 < \rho_3 < \rho_2$$

Q.18 Answer:

40 g more than R

Solution:

@IITJEE_Advanced

Upthrust = weight of $40\ \mathrm{cm}^3$ of water $= 40\ \mathrm{g}$ = down thrust on water

Q.19 Answer:

$\frac{\ell}{\sqrt{2\pi}}$

Solution:

As $A_1v_1 = A_2v_2$ (Principle of continuity) or, $\ell^2\sqrt{2gh} = \pi r^2\sqrt{2g \times 4h}$ (Efflux velocity $= \sqrt{2gh}$)

$\therefore \quad r^2 = \frac{\ell^2}{2\pi}$ **or** $r = \sqrt{\frac{\ell^2}{2\pi}} = \frac{\ell}{\sqrt{2\pi}}$

Q.20 Answer:

07.00

Solution:

For limb of radius $2 \quad mm, h_1 = \frac{2T\cos\theta}{r_1\rho g}$

For limb of radius $1\mathrm{mm}, \mathrm{h}_2 = \frac{2\mathrm{T}\cos\theta}{\mathrm{r}_2\rho\mathrm{g}}$

$\therefore \quad \mathrm{h}_2 - \mathrm{h}_1 = \frac{2\mathrm{T}\cos\theta}{\rho\mathrm{g}}\frac{1}{\mathrm{r}_2} - \frac{1}{\mathrm{r}_1}$

$= \frac{2 \times 0.07 \times 1}{10^3 \times 10}\frac{1}{10^{-3}} - \frac{1}{2 \times 10^{-3}}$

$= 0.0141 - \frac{1}{2}$

$= 0.007\mathrm{m}$

$= 7mm$

@IITJEE_Advanced